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Empirical Analysis of Ambulance Travel Times: The Case of Calgary Emergency Medical Services

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Using administrative data for high-priority calls in Calgary, Alberta, we estimate how ambulance travel times depend on distance. We find that a logarithmic transformation produces symmetric travel-time distributions with heavier tails than those of a normal distribution. Guided by nonparametric estimates of the median and coefficient of variation, we demonstrate that a previously proposed model for mean fire engine travel times is a valid and useful description of median ambulance travel times. We propose a new specification for the coefficient of variation, which decreases with distance. We illustrate how the resulting travel-time distribution model can be used to create probability-of-coverage maps for diagnosis and improvement of system performance.

Key words: ambulance service; probability-of-coverage map; nonparametric; travel

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1. Introduction

Performance measures for such emergency services as medical, fire, and police typically focus on response times. Reaching 90% of urgent urban emergency medical services (EMS) calls in 9 minutes is a common target in North America (Fitch 2005), for example, and the National Health Service in the United Kingdom sets targets of 75% in 8 minutes and 95% in 14 minutes for such EMS calls (Department of Health 2008). Similarly, the U.S. National Fire Protection Association (2004, §5.2.4.1.1) recommends a target of 90% for a fire engine to reach a building fire within four minutes. Accurate prediction of such performance measures requires information about the distribution of the two components of response time: travel time and pretravel delay (from receipt of the call until the dispatched vehicle starts moving). For ambulances, travel time to the scene is often the largest component of the response time. Although response time performance targets for these emergency services correspond to specific quantiles of the response time distribution, it is generally useful to know the entire response time distribution to predict performance measures that are more closely related to patient outcomes (Erkut et al. 2008). Green and Kolesar (2004) provide a recent review of management science applications for improving emergency response.

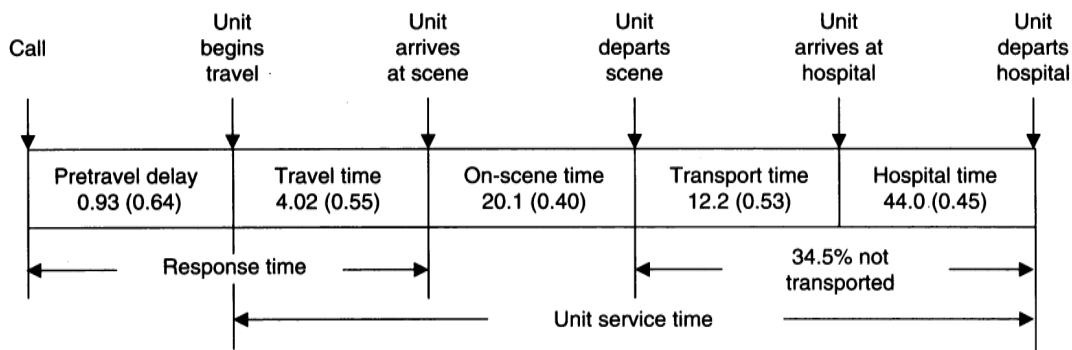
In this paper, we estimate the ambulance travel-time distribution and its dependence on distance. We

use data from the EMS system in Calgary, Alberta, a city of one million inhabitants and a population density of 1,360 people per square kilometer (compared to 2,750 people/km² for Los Angeles, for example). We focus on high-priority calls because (a) long travel times are those most likely to have adverse consequences for high-priority calls, and (b) travel speeds are higher for high-priority calls because of the use of “lights and siren” (Ho and Casey 1998, Hunt et al. 1995), and it is necessary, therefore, to control for call priority. In 2003, the Calgary EMS system responded to 7,457 high-priority calls. We removed 571 outliers using procedures described in the online appendix, §EC.1 (provided in the e-companion),¹ leaving $n = 6,886$ observations. Each observation consists of time stamps (generated by paramedics and dispatchers pressing buttons) for the events illustrated in Figure 1 and coordinates (recorded automatically by global positioning system units in the ambulances) for the call and the ambulance at the time of dispatch. We used the coordinates for each call to compute the shortest travel distance, using the Calgary road network.

Related research has focused on point estimation of travel times when the origin and destination are known. Analysts at the RAND Corporation developed

¹ An electronic companion to this paper is available as part of the online version that can be found at <http://mansci.journal.informs.org/>.

Figure 1 Events and Time Intervals for an EMS Call, with Medians (in Minutes) and Coefficients of Variation for Each Interval



an approach to computing travel-time point estimates (Kolesar 1979) whereby time increases with the square root of the distance for short trips and linearly with the distance for long trips. Kolesar et al. (1975), referred to hereafter as KWH, specify assumptions that imply this two-part functional form and give meaning to its parameters: that the vehicle accelerates from the origin at rate a until it reaches cruising velocity v_c , which is maintained until the unit begins to decelerate (at the same rate a), coming to a stop at the destination (see the “KWH speed” in Figure 2). The distance and time required to achieve cruising velocity are $d_c = v_c^2/(2a)$ and $t_c = v_c/a$, respectively. The travel time T , conditional on distance d (the KWH function), is

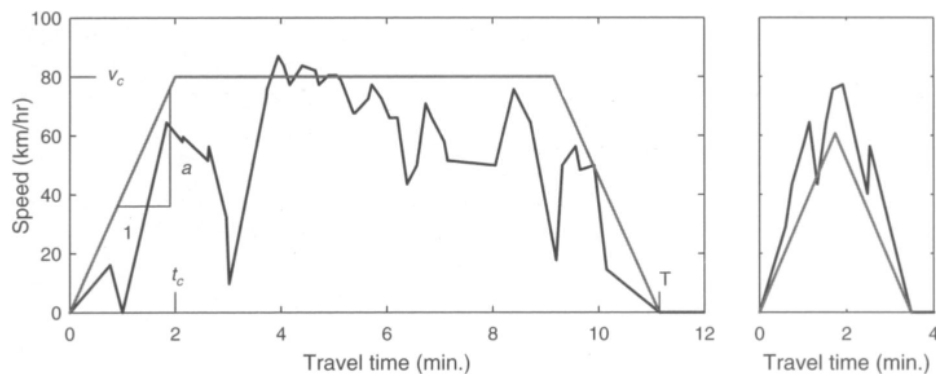
$$\text{median}[T \mid d] \equiv m(d) = \begin{cases} 2\sqrt{d/a} & d \leq 2d_c, \\ v_c/a + d/v_c & d > 2d_c. \end{cases} \quad (1)$$

Kolesar et al. (1975) used this relationship to model the mean travel time. We express the relation using the median rather than the mean because travel times are positive and travel-time distributions are, therefore, likely to be skewed. In contrast to the KWH function, common modeling practice treats distance and travel time interchangeably, which implies a constant speed. Carson and Batta (1990) document how this practice can result in overestimation of EMS system

performance, and Ingolfsson et al. (2003) found that accounting for acceleration/deceleration effects was crucial for EMS simulation model validity.

In reality, a vehicle’s speed profile will not adhere exactly to the “KWH speed” shown in Figure 2 because of traffic lights, stop signs, traffic conditions, weather, or other reasons. The acceleration and deceleration phases should be viewed as corresponding to travel on residential or arterial roads, whereas the cruising speed phase corresponds to highway travel. The validity of the KWH assumptions can be investigated empirically using automatic vehicle locator (AVL) data, which include location (latitude and longitude) for every ambulance, recorded automatically (in Calgary) every minute for stationary vehicles and approximately every 150 meters for moving vehicles. Figure 2, which is based on data from Edmonton, Alberta, shows AVL speeds for two calls with a superimposed visually fitted KWH speed profile—one in which the speed leveled off during the middle portion of the trip (a “long” trip), and another in which the speed did not level off (a “short” trip). Although the AVL speed varies considerably, it appears reasonable to view it as the KWH speed corrupted by noise. Bearing in mind that the primary purpose of the KWH function is to predict total travel time rather than the speed profile, the AVL data support the use

Figure 2 AVL Speed (Black) and Visually Fitted KWH Speed (Gray) for One “Long” Call and One “Short” Call



of the KWH function as a reasonable approximation for median travel times.

Our paper provides the first thorough examination of the distribution of travel times in a way that can be easily incorporated into stochastic models for EMS planning. Unlike the approach in this paper, past approaches often failed to account appropriately for the stochastic nature of travel time, which, even for a given origin and destination, will vary because of route choice, driver characteristics, traffic conditions, geographic aggregation of points into zones, and other known and unknown factors. This paper makes four main contributions. (1) We demonstrate that the KWH model for mean fire engine travel times is a valid and useful description of median ambulance travel times. We perform our validation by comparing the KWH function to the best-fitting nonparametric estimate of the median function; that is, we compare the KWH function to an infinite number of other possible functions, and find that it performs about as well as any of them. (2) We expand the KWH model to represent explicitly the full probability distribution of travel times conditional on travel distances. We propose a statistical model that allows the median travel time and its coefficient of variation (CV) to vary with distance, and we account for skewness and excess kurtosis (i.e., the distribution has a higher peak and fatter tails than a normal distribution). (3) We present and justify a parametric specification for how the CV varies with distance. (4) We show how our model of travel-time distributions can be used to create probability-of-coverage maps that facilitate decisions about where to locate stations and how to allocate ambulances to them.

The implications of this research extend beyond emergency services. Travel-time distributions are also critical in nonemergency applications, such as the routing of pickup and delivery vehicles (see Ratliff and Zhang 1999, Cook and Russell 1978, Kenyon and Morton 2003, Topaloglu 2006). Delivery time is one of the operational dimensions on which private-sector firms compete (e.g., see So 2000). A familiar example is the delivery-time guarantees offered by companies such as Domino's Pizza.

Section 2 of this paper introduces a statistical approach that allows the travel-time median and CV to depend nonparametrically on distance. In §3, we parameterize the dependence on distance, leading to a theoretically appealing and easily implementable travel-time distribution model. In §4, we illustrate the practical use of the travel-time distribution for EMS planning, using probability-of-coverage maps. Section 5 summarizes the contributions of this paper. The online appendix in the e-companion provides further information about the data and the statistical methodology used.

2. Nonparametric Median and CV Functions

In an attempt to find an easily implementable and interpretable model for use in EMS planning, this section discusses a new model for the conditional distribution of travel time, where the travel time median and CV depend nonparametrically on distance. We then use the results of this analysis as statistical guides for the parametric median and CV functions that we propose in §3. In our preliminary analysis, we divided the travel-time observations ($n = 6,886$) into eight one-kilometer distance classes and investigated the travel-time distribution in each distance class. The distributions were highly skewed, but a logarithmic transformation resulted in approximately symmetric distributions. We found that shifted and scaled Student's t distributions provided a good fit to the log-travel-time distributions—considerably better than the fit of normal distributions—indicating that the log-travel-time distributions have heavy tails. Use of the normal distribution resulted in an underestimation of cumulative travel-time probabilities by up to nine percentage points in the upper tail and overestimation by a similar amount in the lower tail (the online appendix, §EC.2, shows the empirical and fitted distributions). In contrast, cumulative probabilities for the t distribution were always within four percentage points of the empirical distribution, reflecting the importance of accounting for heavy tails. Depending on the distance class, the best-fitting degrees of freedom, τ , for the t distributions varied from 3.3 to 5.4 (see Johnson et al. 1994 for information about the t distribution when τ is allowed to be a noninteger).

Guided by this preliminary analysis, we propose a regression model in which the log of the i th travel-time observation T_i follows a t distribution that is shifted by $\ln(m(d_i))$ (d_i is the distance for the i th observation) and scaled by $c(d_i)$:

$$\ln(T_i) = \ln(m(d_i)) + c(d_i)\varepsilon_i, \quad i = 1, \dots, n, \quad (2)$$

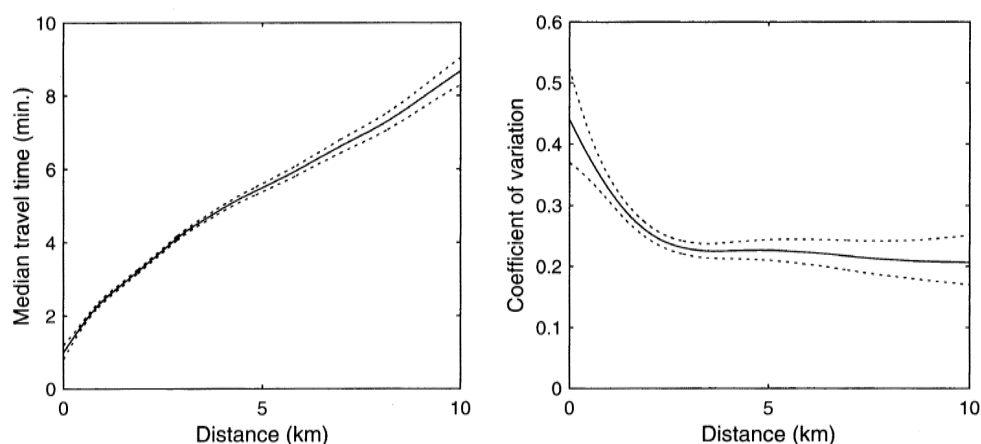
where ε_i follows a centered t distribution with τ degrees of freedom. In this model, we allow the travel-time distributions to vary smoothly with distance (without a need for grouping) via the functions $m(d)$ and $c(d)$. We do not allow τ to vary with distance, because the fitted τ from the preliminary analysis showed little variation between distance classes.

Based upon (2) and $\text{var}[\varepsilon_i] = \tau/(\tau - 2)$ (assuming that $\tau > 2$), the expected value of $\ln(T_i)$ is $\ln(m(d_i))$, and the variance is $c(d_i)^2\tau/(\tau - 2)$. By reversing the log transformation, we obtain

$$T_i = m(d_i) \exp(c(d_i)\varepsilon_i), \quad i = 1, \dots, n, \quad (3)$$

and we see that $m(d_i)$ is the median of T_i . It turns out that $c(d_i)$ is approximately equal to a centile-based

Figure 3 Nonparametric Estimates of the Median and CV Functions with 95% Confidence Limits (Dotted Lines)



CV for T_i (Rigby and Stasinopoulos 2006; see also the online appendix, SEC.3). To simplify the exposition in the remainder of this paper, we refer to $c(d)$ as the CV of T , omitting the qualifiers “approximate” and “centile based.” In addition to fitting the data better, a t distribution for ε has the advantage of robustness over a normal distribution, in that the estimates of the median and CV are less sensitive to outliers (Lange et al. 1989).

Figure 3 shows the fitted median and CV functions with 95% confidence intervals, estimated by the penalized maximum likelihood approach of Stasinopoulos and Rigby (2007). The shape of the estimated median function is similar to the KWH function, with a transition from the square-root shape to the straight line occurring at approximately four kilometers. The estimated CV function decreases, first rapidly and then more slowly, with a transition occurring at approximately the same distance as for the median function. The estimated degrees of freedom is 3.71, confirming the heavy tails observed in the preliminary group analysis. We also examined the possibility of simplifying the model given in (3) without compromising its predictive power, by restricting one or more components. Such simplifications included $m(d)$ being linear and $c(d)$ being constant, as outlined in the online appendix, SEC.4.

After scrutinizing various versions of model (3), we concluded that to capture the characteristics of the travel-time distribution, it is vital to allow both the median and the CV of the travel time to vary nonlinearly with distance. It is also necessary to account for the tails of the travel-time distribution, which are heavier than those of a normal distribution.

3. Parametric Median and CV Functions

In this section, we use theoretically appealing parametric specifications for the median and CV functions

of model (3), resulting in a model that is easier to implement and interpret. We use the KWH specification (1) for the median function, based on its use in similar situations and its close resemblance to the nonparametrically estimated median function. In addition, we use a new specification for the CV function.

Our CV specification is guided by the nonparametric CV estimate that we obtained in §2 and is motivated by the work of Herman and Lam (1974), who studied commuter travel times. They assumed that the travel time for a trip over distance d equaled the sum of independent and identically distributed travel times over each of the d distance units (assume d is an integer), with the unit-distance travel time having mean μ and variance σ^2 . It follows that $E[T|d] = \mu d$ and $\text{var}[T|d] = (\sigma^2/\mu)E[T|d]$. We generalize Herman and Lam’s (1974) assumptions as follows. (1) Under “average conditions,” the travel time T' has $\text{var}[T'|d] = b_0 + b_1 E[T'|d]$. The constant term b_0 represents variability at the beginning and end of a trip that is independent of trip distance and could be caused, for example, by variability in the recording of travel time or spatial aggregation of the locations of origins and destinations, leading to inaccurate distance measurement. Coefficient b_1 corresponds to Herman and Lam’s (1974) σ^2/μ and can be thought of as representing short-term variation in speed within a trip. (2) Some of the variability in T is caused by variation from trip to trip in such factors as the weather or traffic conditions, and we represent this variation with random variable B , with expected value 1 and variance b_2 . We represent the travel time for a randomly chosen trip as $T = BT'$, with B and T' independent. One can interpret b_2 as a measure of variability because of factors that are not included in the model.

With these assumptions, and defining the coefficient of variation in the usual way ($\text{CV}^* = \sqrt{\text{var}[T|d]}/E[T|d]$), it is straightforward to show that

$$\text{CV}^* = \frac{\sqrt{b_0(b_2 + 1) + b_1(b_2 + 1)E[T|d] + b_2E[T|d]^2}}{E[T|d]}.$$

Next, we assume that the functional form of the relationship between CV^* and $E[T | d]$ is the same as for the relationship between $c(d)$ and $m(d)$, that is, we parameterize $c(d)$ as follows:

$$c(d) = \frac{\sqrt{b_0(b_2 + 1) + b_1(b_2 + 1)m(d) + b_2m(d)^2}}{m(d)}. \quad (4)$$

With this specification, $c(d)$ approaches infinity as $d \rightarrow 0$ and the positive constant $\sqrt{b_2}$ as $d \rightarrow \infty$, consistent with the shape observed in Figure 3 and with data reported in Herman and Lam (1974).

We estimated the model components ($m(d)$, $c(d)$, and τ) via the likelihood approach, using Microsoft Excel's Solver to maximize the log likelihood. The estimated parameters were $a = 41.0$ km/hr/min, $v_c = 100.7$ km/hr (which implies $2d_c = 4.13$ km), $b_0 = 0.336$, $b_1 = 0.000058$, $b_2 = 0.0388$, and $\tau = 4$ (we restricted τ to be integer). The 100.7 km/hr cruising speed is reasonable, because it is slightly higher than the 90 km/hr speed limit on freeways within the City of Calgary, and it is likely that trips longer than 4.13 kilometers will involve at least some free-way travel. The resulting travel-time model is

$$\begin{aligned} T &= \hat{m}(d) \exp(\hat{c}(d)\varepsilon) \quad \text{with } \hat{\tau} = 4; \\ \hat{m}(d) &= \begin{cases} 2.42 \sqrt{d} & d \leq 4.13 \text{ km}, \\ 2.46 + 0.596 d & d > 4.13 \text{ km}; \end{cases} \quad (5) \\ \hat{c}(d) &= \frac{\sqrt{0.349 + 0.000060 \times m(d) + 0.0388 \times m(d)^2}}{m(d)}. \end{aligned}$$

Figure 4 shows that the parametric estimates of the median and CV functions closely resemble their nonparametric counterparts (Figure 3) and are generally within the nonparametric 95% confidence limits, except for extremely short distances. The two median estimates (parametric and nonparametric) can also be compared using an R^2 -like measure (Koenker and Machado 1999), which we define as $R^2 = 1 - \sum_i |T_i - \hat{m}(d_i)| / \sum_i |T_i - \hat{m}|$, where $\hat{m}$ is the median of all of the travel times in the sample.² The R^2 values were 35.53% and 35.77% for the parametric and nonparametric models, respectively, reflecting the similarities observed in the graphs of the median estimates. Overall, the simplified model in (5) with parametric functions $m(d)$ and $c(d)$ captures the important features of the travel time distribution almost as well as model (3), which does not restrict

the forms of the median and CV functions. (The online appendix, §§EC.4 and EC.5, provides computational details and further analysis of the predictive power of the models.)

To validate the parametric model (model (5)) further, we assessed its out-of-sample accuracy using a cross-validation approach whereby we estimated the model parameters from a randomly chosen "training sample" of 689 observations (10% of the entire sample) and validated in the "holdout sample" (the remaining 90% of the entire sample). Our choice of validation criterion was guided by commonly used EMS performance measures, which focus on the probability that response times are less than some standard. A typical standard is eight minutes, but the response time consists not only of travel time but also of the pretravel delay, which typically averages one to three minutes; thus the "budget" for travel time is typically between five and seven minutes. Therefore, our validation focused on how accurately the proposed model estimates $\Pr\{T \leq x | D = d\}$, where $x = 5, 6$, and 7 minutes.

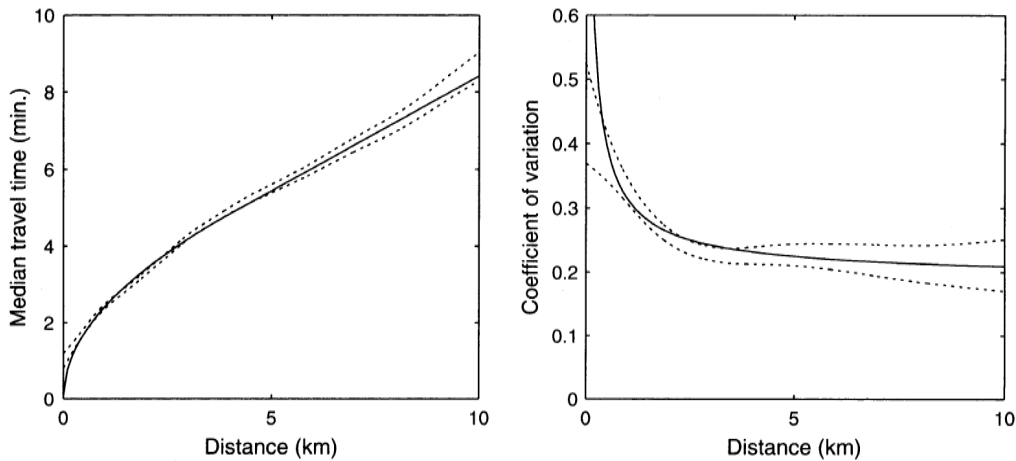
To calculate the "target" probabilities for validation, we grouped the holdout sample using 0.5-kilometer distance intervals; for each interval, we computed the empirical probability that the travel time was less than or equal to five, six, or seven minutes. We did this for 50 randomly chosen training samples. Figure 5 shows results for a representative training sample. (Online appendix, §EC.6, contains a discussion of how we chose this training sample from the 50 that we generated.) The right panel of Figure 5 compares the target and fitted cumulative probabilities. We see that the accuracy of the probabilities decreases with distance, which is expected, because the sample sizes are smaller for the groups corresponding to larger distances. Accuracy is more important for shorter distances, however, because most EMS travel distances are relatively short and there is a low probability of meeting response time standards when travel distance is long. For distances below four kilometers, we see that the fitted travel-time probabilities are generally within two percentage points of the empirical probabilities. We also assessed the validity of the parametric median and CV functions. We calculated the target medians and CVs via their empirical counterparts using the grouped holdout sample. The left panel of Figure 5 shows excellent agreement between the fitted and target medians. The comparison between the fitted and target CVs (not shown) was similar to the fit for the $\Pr\{T \leq x | D = d\}$ measures.

4. Implications for Emergency Systems Planning

To illustrate the practical use of model (5), we outline how to create a "probability-of-coverage map," as

² This is an adaptation of the usual R^2 , i.e., $1 - \sum_i (T_i - \hat{m}(d_i))^2 / \sum_i (T_i - \hat{m})^2$, to the context of medians. The median minimizes the sum of absolute errors, not the sum of squared errors, which is the case for means. We also computed R^2 using the usual definition, and it was about seven percentage points higher than the median-based R^2 .

Figure 4 Parametric Estimates of the Median and CV Functions with Nonparametric 95% Confidence Limits (Dotted Lines)



shown in Figure 6. The maps in the figure show the probability that an ambulance will reach high-priority calls at various locations in the City of Calgary within eight minutes.

The response time for an EMS call equals the sum of pretravel delay L and travel time T . Historical data indicate that for high-priority calls, the distribution of L (measured in seconds) has a discrete mass of 0.0055 at zero, and the nonzero part of the mixture is nicely fitted by $L = \exp(\mu + \sigma \varepsilon)$ with $\mu = 4.68$, $\sigma = 0.38$, and $\tau = 5.37$. Furthermore, L and T are approximately independent. We estimate the probability of coverage for location k as $\Pr\{\text{response time to } k \leq 8\} = \sum_j p_{kj} \Pr\{L + T(d_{kj}) < 8\}$, where p_{kj} (the probability that the ambulance that responds is from station j), is computed using the version of the approximate hypercube model described in Budge et al. (2009) (alternatively, discrete event simulation could be used for more accurate estimation of the “dispatch probabilities” p_{kj}). The distance d_{kj} from station j to location k

is the shortest road-network distance. We used a random variate generator for the t distribution to simulate L and $T(d_{kj})$ for each location–station pair, to estimate the coverage probability $\Pr\{L + T(d_{kj}) < 8\}$ to within 0.01, with 95% confidence (this required 2,500 replications).

Map (a) in Figure 6 shows the probability of coverage, assuming that the closest available ambulance responds to every call. This map shows areas with low probability of coverage in the northeast and northwest regions of the city. Increasing coverage in these areas would require either the deployment of additional ambulances to nearby stations or the building of new stations. To determine the more appropriate option, one can regenerate the map under the assumption that all calls receive a response from the closest station (by setting $p_{kj} = 1$ for the station j that is closest to location k). Map (b) was generated using this assumption, and suggests that coverage in the northwest region can be improved considerably by

Figure 5 Parametric Estimates of the Median and the Probability of a Travel Time Less Than Five, Six, or Seven Minutes, Based on a 689-Observation Training Sample, Compared with Empirical Estimates From a Holdout Sample

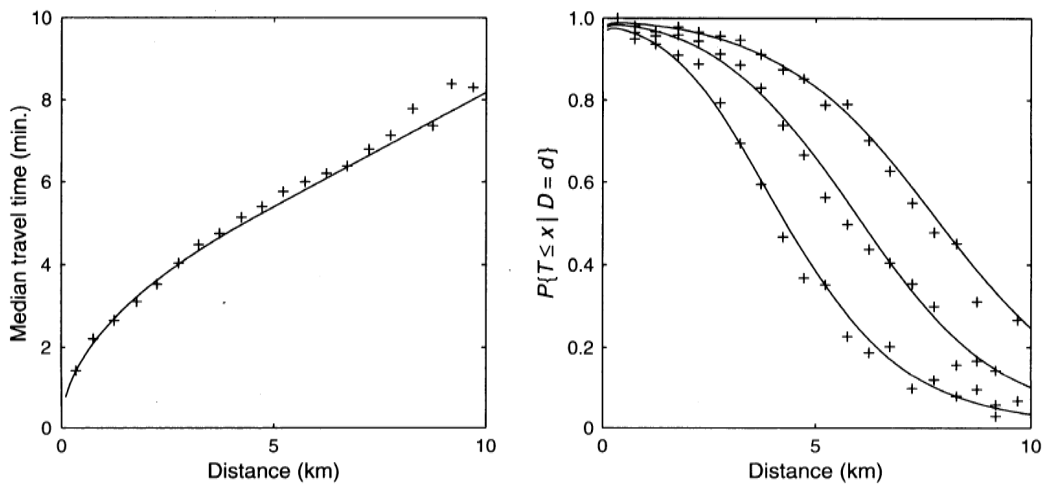
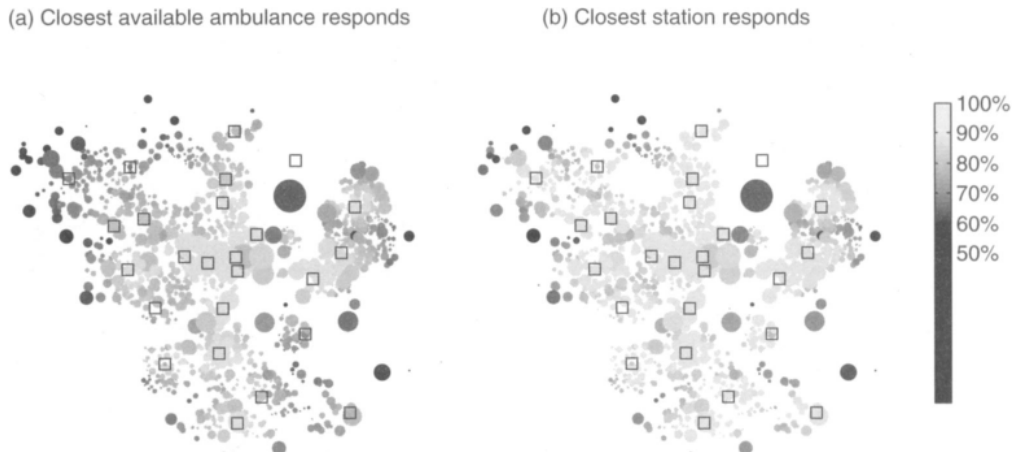


Figure 6 Probability of Coverage Map

Notes. Locations of ambulance stations are depicted as black squares. The colors of the other locations indicate the probability of coverage.

deploying more ambulances to existing stations in that region. In the northeast region, however, it may be necessary to add stations before performance can be improved.

EMS operators have found probability-of-coverage maps easy to understand, and sometimes, with no further analysis, the maps themselves suggested operational improvements. In one instance, after seeing a probability-of-coverage map, an EMS operator recommended to the city council that retirement homes not be built in a particular area that had low EMS coverage probability. In other situations, we have used probability-of-coverage maps to illustrate the likely consequences of adding new stations, as recommended by an optimal station location model. It is common practice in EMS planning to develop maps that show areas of coverage around each station, based on average travel-time estimates. Practitioners have told us that they find probability-of-coverage maps to be a more effective display of system performance.

The approach we have outlined here could be useful to inform decisions in a variety of delivery contexts, in which response and delivery time, and delivery-time guarantees, are important.

5. Summary

In this paper, we revisited the modeling and statistical analysis of EMS vehicle travel times and how travel time varies with distance. We used an approach that allows the relationship between median travel time and distance to be any smooth curve. We confirmed the importance of going beyond a constant-average-speed assumption to model travel times accurately, and demonstrated that a model that was previously proposed for mean fire engine travel times provides an excellent fit to median ambulance travel times. Going beyond previous work, we investigated the entire distribution of travel time conditional on

distance, and demonstrated that these conditional distributions could be made symmetric with a log transformation. Furthermore, we demonstrated that to fit these distributions accurately, it was important to account for tails that are heavier than those of a normal distribution and a coefficient of variation that decreases with distance. Guided by previous work, nonparametric analysis, and our experience with EMS systems, we proposed and validated a simple model that accurately captures the important features of the travel-time data. We illustrated the use of the proposed model to construct probability-of-coverage maps, which we have found useful for model validation and for summarizing and diagnosing EMS system performance.

6. Electronic Companion

An electronic companion to this paper is available as part of the online version that can be found at <http://mansci.journal.informs.org/>.

Acknowledgments

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